SURAJKUMAR KADIYAM BALAJI

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EDUCATION

Northeastern University - Boston, MA **GPA : 4/4**
MS in Electrical and Computer Engineering (Microsystems Materials and Devices). **Dec 2026 (Expected)**
Coursework : Computer Architecture, VLSI Design, Electronic Materials, Solid State Devices, Analog IC, Wireless and IoT

Anna University - Chennai, India **GPA : 3.70/4**
BE in Electrical and Electronics Engineering (University Merit Scholar) **Sept 2018 – Jun 2022**
Coursework : Power Electronics, Embedded Systems, Digital Signal processing, Power System Operation and Control

WORK EXPERIENCE

SharkNinja – Needham, MA **Jan 2026 – Jun 2026**
Electrical Engineering Coop – Robotics Team

- Built an automated log-analysis dashboard using **Python** for RV2900 robot vacuums, cutting fault investigation time from **2–4 hours** to **minutes** with **100% fault recall**.
- Designed **capacitive-sensing** (**cap sense**) boards for floor detection in **Altium**, developing schematics and layouts for reliable surface sensing.

Northeastern University – Boston, MA **Jan 2025 – Dec 2025**
Teaching Assistant – Digital Design and Computer Architecture

- Guide students in **FPGA**-based digital design labs by troubleshooting circuit implementation and debugging in **Vivado** and **Quartus**, enabling smoother lab progress and timely completion of **10+ weekly experiments**.
- Developed **SystemVerilog/UVM** testbenches with assertions & coverage models; automated log parsing using **Python**, reducing debug effort by **25%**.
- Reviewed microarchitecture-level projects (**pipelining, cache, branch prediction**) to align with industry-style **CPU/SOC verification** practices.

Tata Motors – Pune, India **Jan 2023 – Jun 2024**
Electrical Engineer – Tata Motors Commercial Vehicles Team

- Designed and validated **embedded ECU subsystems** with emphasis on **power distribution**, **analog/digital interfaces**, and **EMI/EMC compliance**, ensuring robust performance across multiple vehicle platforms.
- Developed **Linux**-based diagnostic scripts in **Python/Bash** to test **ECU communication (CAN/I^2C/SPI/USB)**, reducing debug cycle time by **22%**.

Skill-Lync E-learning Company – Chennai, India **Jul 2022 – Dec 2022**
Embedded Engineer Intern – Software Verification and Validation, AVR Bare Metal Team

- Built and tested **PCB prototypes** by integrating **analog front-end circuits** with **digital controllers**, supporting **embedded system bring-up** and validation.
- Developed **bare-metal verification flows** for **AVR microcontrollers** using **JTAG**, coverage metrics, and debug scripts to improve debugging efficiency.

PROJECTS

Functional Verification of HTAX Cross-bar using System Verilog & UVM **Mar 2025**

- Built a **UVM** environment and applied **coverage-driven/formal methods** to validate **HTAX crossbar**.
- Automated **log extraction and coverage analysis** in **Cadence IMC** using **Python**, reducing manual effort by **~30%**.
- Achieved **98% functional coverage** and integrated results into **Cadence VManager** for streamlined tracking.

Design and Implementation of 32-bit ALU in 45nm using Cadence **Mar 2025**

- Designed a 32-bit **ALU** in **180nm CMOS** using **Cadence Virtuoso** by scaling custom 1-bit ALU blocks.
- Applied **Back Gate Forward Substrate Biasing** to reduce delay by **~40%** and improve switching speed.
- Conducted **timing and power analysis** across different **PVT** corners to validate reliability.
- Verified **ALU** performance via **Spectre** simulations using hierarchical schematics and testbenches.

Branch Prediction and Cache Design Simulator **Oct 2024**

- Implemented adaptive **branch predictors** in **Python**, improving accuracy by **26%** on **SPECint2000** benchmarks.
- Built and tuned a **cache simulator** in **DineroIV**, optimizing policies and prefetching for a **15% increase in hit rate**.
- Characterized cache predictor performance across **block sizes**, **pipeline flush penalties** to evaluate **IPC trade-offs**.

SKILLS

Coding Languages : Python, C, Embedded C, Verilog, System Verilog, UVM, Assembly Language, Linux
Software : Cadence Virtuoso, Xcelium, Verisium, Quartus Prime, STM32, AutoCAD, KiCAD, TCAD, OrCAD, VESTA, HSPICE, PSPICE, Matlab, Xilinx ISE
Certification : System Verilog for Design and Verification, Embedded C essentials, Software Verification and Validation